UCEED 2026

Undergraduate Common Entrance Examination for Design — MOCK QUESTION PAPER

Paper Specific Instructions

1. Total duration: 3 hours. The paper contains Part A and Part B. Part A: 2 hours; Part B: 1 hour.

2. Part A appears on the computer for the first 2 hours. Answers to Part A must be entered on the computer within that time. Part B appears after Part A and answers must be written in the answer book provided by the invigilator.

3. Part B contains two questions of 50 marks each. Both are compulsory. No negative marking in Part B.

Marking Scheme — Part A

Section 1 (NAT): 14 questions × 4 marks = 56. No negative marks.

Section 2 (MSQ): 15 questions. Partial marking as per instructions; incorrect selections may incur −1.

Section 3 (MCQ): 28 questions × 3 marks = 84. Incorrect answer: −0.71 marks.

Part A — Section 1: Numerical Answer Type (NAT) — Questions 1–14 (4 marks each)

Q.01

A digital clock displays time in HH:MM (00:00–23:59). How many times are all four digits prime numbers (2,3,5,7)?

Q.02

Eight identical small cubes form a 2×2×2 cube. Each small cube has exactly two faces painted each of red, blue, and green. The assembled large cube shows only red on the top face, only blue on the bottom face, and only green on all four side faces. How many small-cube faces are hidden inside the large cube?

Q.03

A triangular pattern: layer 1 has 1 up triangle; layer 2 has 3 (2 down,1 up); layer 3 has 5 (3 up,2 down). Continuing for 7 layers, how many triangles point upward?

Q.04

A spiral staircase has 15 steps; each step rotates 24° from the previous. Starting at step 1 facing North, after reaching step 15, what direction (° clockwise from North) is faced?

Q.05

Five transparent sheets (Red, Blue, Yellow, Green, Magenta) are stacked. Given pairwise color mixing (Red+Blue=Purple; Blue+Yellow=Green; Red+Yellow=Orange; Red+Green=Brown) and final visible color Brown with Blue fixed as 3rd from top, how many stacking arrangements are possible?

Q.06

A hexagonal table (diameter 180 cm) has 6 chairs at vertices. Six people of heights 150,155,160,165,170,175 cm sit so each can see over shoulders of two opposite people. What is the minimum height difference (cm) required between a person and the person directly opposite?

Q.07

Count right/down paths from A to B that spell "DESIGN" when read along the path on the grid:

D E S I

E S I G

S I G N

Q.08

Potter's wheel rotates at 120 RPM. A point 8 cm from center experiences pressure for exactly 3 rotations. What is total distance travelled by contact point? (Use π=3.14)

Q.09

Gears A (48 teeth, 60 RPM clockwise), B (36 teeth), C (24 teeth) in series. If A completes 10 rotations, how many rotations does C complete?

Q.10

Nested golden-ratio rectangles: outermost 161.8 × 100 cm (ratio ≈1.618). Remove a square from the longer side successively 4 times. Area of the 5th rectangle? (Round to nearest integer)

Q.11

Bookshelf pieces: 30, 45, 60 cm. Stack to total 240 cm tall using at least one of each piece. What is the minimum total number of pieces?

Q.12

A cube with edge 10 cm has each vertex truncated by cutting 2 cm along each of the three incident edges. What is the total number of edges in the resulting polyhedron?

Q.13

At 3:15 the angle between hour and minute hands is X°. Between 4:00 and 5:00, at what time is the angle exactly 2X°?

Q.14

A tessellation of regular hexagons and equilateral triangles has 8 fully visible hexagons; triangles fill all gaps and surround hexagons. How many triangles are needed?

Part A — Section 2: Multiple Select Questions (MSQ) — Questions 15–29

Q.15

Q.15

Which shadow patterns are impossible with a single solid 3D object under a single point light source? (Select all that apply.)

A. A circle that transitions to a square

B. An oval that transitions to a rectangle

C. A shadow with two completely separate disconnected parts

D. A shadow that changes size but not shape

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Q.16

A Möbius strip is cut along its centerline. Which statements are TRUE? (Select all that apply.) A. Results in two separate strips B. Results in one longer strip with two full twists C. Resulting strip(s) have twice the surface area of original D. The cut can never reconnect to form a closed loop

Q.17

Which Indian design elements share the same cultural origin region? (Select all that apply.) A. Warli painting and Paithani saree B. Madhubani art and Phulkari embroidery C. Pattachitra and Ikat weaving D. Kalamkari and Kondapalli toys

Q.18

Perspective drawing (single vanishing point). Which statements are TRUE? (Select all that apply.) A. Horizontal lines parallel to picture plane remain parallel B. Vertical lines always converge at the vanishing point C. Objects closer to vanishing point appear smaller D. Horizon line always passes through center of canvas

Q.19

Which processes involve chemical change? (Select all that apply.) A. Electroplating metal B. Vacuum forming plastic C. Vulcanizing rubber D. Anodizing aluminum

Q.20

A half-filled transparent sphere on white surface under sunlight. Which phenomena can be observed? (Select all that apply.) A. Refraction creating a focal point B. Total internal reflection at certain angles C. Dispersion creating a spectrum D. Shadow will be completely circular

Q.21

Which letter combinations have only rotational symmetry and NO reflective symmetry? (Select all that apply.) A. NON B. SOS C. ZOO D. HID

Q.22

Five identical springs: pulling one spring extends 10 cm under 5 N. Which statements are TRUE? (Select all that apply.) A. Two springs in parallel extend 5 cm under 5 N B. Two springs in series extend 20 cm under 5 N C. Three springs in series extend 15 cm under 10 N D. Three springs in parallel require 15 N to extend 10 cm

Q.23

Which designers/architects are associated with the Bauhaus movement? (Select all that apply.) A. Ludwig Mies van der Rohe B. Le Corbusier C. Marcel Breuer D. Frank Lloyd Wright

Q.24

Statements about complementary colors. Which are TRUE? (Select all that apply.) A. When mixed equally they create gray/brown B. They are opposite on the color wheel C. They create maximum contrast when adjacent D. Red and blue are complementary

Q.25

A regular octagonal plan footprint — which elevation views are possible? (Select all that apply.) A. A rectangle with a triangular roof B. A perfect octagon C. Two rectangles at different heights D. A trapezium with parallel top and bottom

Q.26

Which materials have lower carbon footprint than Portland cement? (Select all that apply.) A. Fly-ash based geopolymer B. Hempcrete C. Cross-laminated timber D. Aluminum composite panels

Q.27

Digital image formats — which statements are TRUE? (Select all that apply.) A. JPEG uses lossy compression B. PNG supports transparency C. GIF can have max 256 colors per frame D. SVG files become pixelated when enlarged

Q.28

Traditional Indian instruments with resonating chambers (gourds/hollow bodies). (Select all that apply.) A. Veena B. Tabla C. Sarangi D. Shehnai

Q.29

Which 3D objects match all four orthographic views (Front, Top, Right, Left)? (Select all that apply.) A. Cylinder with rectangular protrusion on one side B. Rectangular prism with a cylindrical hole through it C. Cone with tip cut at an angle D. Hemisphere attached to a cube face

Part A — Section 3: Multiple Choice Questions (MCQ) — Questions 30–57

Q.30

If a regular icosahedron is placed with one vertex touching a table, what is the maximum number of edges visible from directly above? A. 10 B. 15 C. 20 D. 25

Q.31

The word "OPTIMIZATION" is written vertically, reflected about a horizontal axis and then rotated 90° clockwise. Which letter is at the bottom? A. O B. N C. I D. T

Q.32

Six people A–F sit around a circle. Given: A opposite D; B two places to the right of A; C between E and F. Who sits opposite C? A. B B. E C. F D. Cannot be determined

Q.33

In traditional bookbinding, which stitching pattern provides the strongest binding? A. Pamphlet stitch B. Coptic binding C. Japanese stab D. Perfect binding

Q.34

A square folded diagonally three times, two small triangular notches cut from folded edges — when unfolded, which shape appears? A. Square with circular pattern B. Eight-pointed star C. Snowflake-like pattern with 8-fold symmetry D. Four separate diamond shapes

Q.35

Gestalt principle where humans perceive complete image despite missing parts: A. Proximity B. Closure C. Similarity D. Continuity

Q.36

Which color space is designed for printing and uses four channels? A. RGB B. HSV C. CMYK D. LAB

Q.37

A 200 m train at 72 km/h is visible at the tunnel entrance for 15 s. What is the tunnel length? A. 100 m B. 200 m C. 300 m D. 400 m

Q.38

Which material has the highest tensile strength? A. Spider silk B. Steel cable C. Kevlar D. Carbon fiber

Q.39

The Ishihara test primarily tests for: A. Visual acuity B. Color blindness C. Depth perception D. Pattern recognition

Q.40

In UI/UX, "affordance" means: A. Cost B. Perceived quality that suggests use C. Load time D. Color contrast ratio

Q.41

Map scale 1 cm = 50 km. Distance 7.5 cm. Car speed 75 km/h. Travel time? A. 3 h B. 4 h C. 5 h D. 6 h

Q.42

Which UNESCO site is famed for chattris and jaali work? A. Khajuraho B. Taj Mahal C. Hampi D. Konark

Q.43

In typography, spacing between two specific letters is adjusted using: A. Leading B. Tracking C.

Kerning D. Baseline

Q.44

Most suitable production method for 100,000 identical plastic bottles: A. Vacuum forming B. Injection molding C. Rotational molding D. 3D printing

Q.45

Fibonacci: ratio of 10th to 9th term (rounded to 2 decimals). A. 1.58 B. 1.62 C. 1.68 D. 1.72

Q.46

Sustainable principle for easy disassembly: A. Cradle to Cradle B. Design for Disassembly C. Biomimicry D. Life Cycle Assessment

Q.47

Sequence: 2,6,12,20,30, ? A. 40 B. 42 C. 44 D. 48

Q.48

Qutub Minar is located in: A. Agra B. Delhi C. Lucknow D. Ahmedabad

Q.49

Non-destructive editing in 3D modeling refers to: A. Editing that doesn't damage the computer B. Ability to modify earlier operation parameters without starting over C. Preserves original file quality D. Collaborative reversible editing

Q.50

Cone, cylinder, hemisphere — same base radius 7 cm and height 14 cm. What is ratio of their volumes? A. 1:2:3 B. 1:3:2 C. 2:3:4 D. 1:4:2

Q.51

Chronological order of art movements: A. Impressionism → Cubism → Art Deco → Pop Art B. Cubism → Impressionism → Pop Art → Art Deco C. Art Deco → Impressionism → Cubism → Pop Art D. Impressionism → Art Deco → Cubism → Pop Art

Q.52

In sections, line type representing objects above cutting plane: A. Solid thick B. Dashed C. Chain D. Dotted

Q.53

Rule of thirds divides the frame into: A. Three horizontal sections B. Three vertical sections C. 3×3 grid with 9 equal parts D. Three concentric circles

Q.54

"Less is More" philosophy designer: A. Frank Gehry B. Ludwig Mies van der Rohe C. Zaha Hadid D. Antoni Gaudí

Q.55

If DESIGN → EFTJHO, then CREATIVE → ? A. DSFBUJWF B. DSFBUJWF C. DQFBSJWF D. DSFBUJWG

Q.56

Heating metal then cooling slowly to reduce hardness and increase ductility: A. Tempering B. Quenching C. Annealing D. Case hardening

Q.57

Waste hierarchy order (most to least preferred): A. Reduce → Recycle → Reuse → Recover → Dispose B. Reduce → Reuse → Recycle → Recover → Dispose C. Reuse → Reduce → Recycle → Recover → Dispose D. Recycle → Reduce → Reuse → Dispose → Recover

Part B — 60 minutes — 100 marks

Q.58 — Drawing (50 marks)

Scenario: 6:30 AM winter morning in a community park. Sun rising with long shadows. Elements: jogger's track, playground, benches, trees, elderly man practicing Tai Chi, three morning walkers on a bench having chai, a young woman jogging with her dog, a street vendor setting up a juice cart, distant apartment silhouettes in mist.

Instructions: Draw the scene from perspective of someone sitting on a bench near the playground. Use pencil only. Show light and shadow consistent with morning sun.

Q.59 — Design Aptitude (50 marks)

Design brief: Create a modular indoor/balcony planter system for urban millennials (25–35 years). Requirements: accommodates 4–6 plants; footprint ≤ 60 cm × 40 cm; self-watering mechanism; tool-free assembly; sustainable/recyclable materials.

Deliverables (visuals only, pencil): 1) Assembled view (isometric/perspective) 2) Section view showing self-watering mechanism with labels 3) Exploded view of modular components 4) One alternative configuration demonstrating modularity. Show key dimensions. No paragraphs of explanation; use short labels only.

End of Question Paper — Notes to Candidates

• This is a mock paper for practice. The actual UCEED paper may vary but follows similar format and marking scheme.

• Calculators, mobile phones, watches and electronic gadgets are not allowed. Blank sheets provided for rough work.